AR-MULTI++ Web:Context-Aware Multilingual Scene Text Recognition, Translation, and Voice Output with Augmented Reality Simulation

A Project Report

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CERTIFICATE

This is to certify that the project report entitled **“AR-MULTI++ Web:Context-Aware Multilingual Scene Text Recognition, Translation, and Voice Output with Augmented Reality Simulation”** is a bonafide record of work carried out by **M. NAVYA (22481A05E1), P.VARSHITHA (22481A05J4), N. KARTHIK (22481A05G9), M.RAJ VARDHAN (22481A05E7)** under the guidance and supervision in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering of Jawaharlal Nehru Technological University Kakinada, Kakinada during the academic year 2025-26.

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LIST OF ABBREVATIONS

Abbreviation	Explanation
AR	Augmented Reality
OCR	Optical Character Recognition
TTS	Text-to-Speech
NLP	Natural Language Processing
UI	User Interface
API	Application Programming Interface
ML	Machine Learning
STT	Speech-to-Text

LIST OF FIGURES

ABSTRACT

In the present era of digital transformation and globalization, multilingual textual information appears frequently in everyday environments such as public signboards, documents, product labels, educational materials, and digital displays. However, understanding such information becomes difficult when the text is written in unfamiliar languages. Traditional translation systems mainly focus on translating typed text and often fail to handle text embedded within real-world images. Furthermore, many existing solutions lack contextual visualization and integrated audio support, limiting their effectiveness in practical scenarios such as travel assistance, education, and accessibility support.

The objective of this project is to develop a robust web-based system capable of recognizing, translating, and presenting multilingual text from real-world images. The proposed system, AR-MULTI++ Web, integrates multiple technologies including Optical Character Recognition (OCR), machine translation, text-to-speech synthesis, and augmented reality-style visual overlay to provide a comprehensive multilingual assistance platform. The system allows users to upload images containing textual information, automatically extracts the text using a cloud-based OCR engine, translates it into a selected target language, and generates corresponding audio output for improved accessibility.

In addition to translation and speech generation, the system overlays the translated text onto the original image while preserving its spatial context. This augmented visualization helps users understand the translated content in relation to its original placement within the scene. The system is implemented using a modular client-server architecture with modern web technologies to ensure scalability, efficiency, and real-time processing capability.

Experimental results demonstrate that the proposed system achieves high accuracy in text recognition, efficient translation performance, and reliable audio generation. By integrating multilingual text recognition, translation, audio synthesis, and contextual visualization into a single platform, the AR-MULTI++ Web system provides an effective solution for overcoming language barriers in real-world environments and enhancing accessibility for users worldwide.

Keywords – Multilingual Text Recognition, Optical Character Recognition (OCR), Scene Text Translation, Augmented Reality Overlay, Machine Translation, Text-to-Speech (TTS), Web-based System, Image Processing.

CHAPTER – 1

INTRODUCTION

1.1 Introduction

In today's rapidly evolving digital world, the presence of multilingual textual information has become increasingly common in everyday environments. Text written in different languages can be found on public signboards, transportation systems, product labels, restaurant menus, educational materials, and digital displays. While globalization has increased connectivity among people from different linguistic backgrounds, language barriers continue to pose significant challenges in understanding such information. Individuals traveling to foreign countries, students accessing global educational resources, and people encountering unfamiliar languages in daily life often struggle to interpret textual content embedded in images or real-world scenes.

With the advancement of computer vision and artificial intelligence technologies, automated text recognition and translation systems have gained considerable attention. Optical Character Recognition (OCR) technology has made it possible to extract textual information from digital images, while machine translation systems enable the conversion of text from one language to another. However, many existing translation tools primarily focus on text-based input and are not designed to effectively interpret text present in natural scene images. Real-world images often contain variations in lighting conditions, font styles, orientations, and complex backgrounds, making text extraction a challenging task for traditional OCR systems.

Another limitation of existing translation systems is the lack of contextual visualization and accessibility support. Many applications translate extracted text but fail to preserve the spatial relationship between the translated text and its original position in the image. As a result, users may struggle to understand the translated content within the context of the original scene. In addition, most conventional systems do not provide integrated audio output, which limits their usability for individuals with visual impairments or reading difficulties.

To address these challenges, there is a growing need for intelligent systems capable of recognizing, translating, and presenting multilingual text directly from real-world images while maintaining contextual understanding. The integration of multiple technologies such as Optical Character Recognition, machine translation, text-to-speech synthesis, and augmented reality visualization can significantly enhance the effectiveness of such systems.

By combining these technologies into a unified platform, it becomes possible to create a seamless user experience that not only translates text but also presents it in a meaningful and accessible manner.

The AR-MULTI++ Web system is designed to provide a comprehensive solution for multilingual scene text recognition and translation. The system allows users to upload images containing textual information, automatically extracts the text using advanced OCR techniques, translates it into a selected target language, and generates corresponding speech output. Furthermore, the translated text is visually overlaid on the original image in an augmented reality style, preserving the spatial context of the text and making the translation easier to interpret.

The development of such systems has significant implications across various domains. In tourism, travelers can quickly understand foreign-language signboards, menus, and instructions. In education, students can access and comprehend learning materials written in different languages. In accessibility applications, visually impaired users can listen to translated text through speech synthesis. Additionally, multilingual text recognition systems can support global communication by reducing language barriers and promoting inclusive access to information.

Despite the progress made in OCR and translation technologies, the integration of these components into a single web-based system capable of real-time processing remains an ongoing research challenge. The AR-MULTI++ Web framework aims to bridge this gap by providing a scalable and user-friendly platform that combines text recognition, translation, audio generation, and augmented visualization within a unified architecture. By leveraging modern web technologies and cloud-based services, the system ensures efficient processing, improved accuracy, and enhanced usability.

Therefore, the AR-MULTI++ Web system represents an important step toward enabling seamless multilingual communication and improving accessibility to textual information in real-world environments. By overcoming language barriers and enhancing contextual understanding, such systems can contribute significantly to the development of intelligent and inclusive digital solutions for the modern world.

1.2 Objectives of the Project:

The primary aim of this project is to develop an efficient system capable of recognizing and translating multilingual text present in real-world images. Traditional translation systems mainly focus on typed text and often fail to interpret text embedded within natural scene images due to variations in font styles, lighting conditions, and complex backgrounds. To overcome these challenges, the proposed AR-MULTI++ Web system integrates technologies such as Optical Character Recognition (OCR), machine translation, text-to-speech synthesis, and augmented reality-based visualization.

The system allows users to upload images containing multilingual text, automatically extracts the text using OCR techniques, and translates it into a user-selected target language. In addition to translation, the system generates audio output through text-to-speech technology, enabling users to listen to the translated content and improving accessibility for visually impaired individuals.

By combining these technologies into a single web-based platform, the project aims to provide a user-friendly solution for overcoming language barriers and improving multilingual text understanding in real-world environments such as travel, education, and daily communication.

1.3 Problem Statement:

Understanding multilingual text present in real-world images remains a significant challenge in many practical applications. Traditional translation systems primarily focus on translating typed or document-based text and often fail to accurately interpret textual information embedded in natural scene images. Variations in font styles, lighting conditions, orientations, and complex backgrounds make it difficult for conventional Optical Character Recognition (OCR) systems to reliably extract text from such images. As a result, users frequently encounter difficulties in understanding foreign-language signboards, documents, and visual information in everyday environments.

Furthermore, most existing translation tools operate independently without preserving the contextual placement of translated text within the original image. This lack of contextual visualization can reduce the clarity and usability of translated results.

The problem becomes even more challenging when multilingual text appears in real-time scenarios such as travel, education, and public information systems. The absence of a unified platform that integrates text recognition, translation, audio generation, and contextual visualization restricts the effectiveness of current solutions. Therefore, there is a need for a comprehensive system capable of accurately recognizing multilingual text from images, translating it into a desired language, generating speech output, and presenting the translated information in a visually meaningful manner.

The proposed AR-MULTI++ Web system aims to address these challenges by integrating Optical Character Recognition, machine translation, text-to-speech synthesis, and augmented reality-style visual overlay into a single web-based platform. By doing so, the system seeks to improve multilingual text understanding, enhance accessibility, and provide a practical solution for overcoming language barriers in real-world environments.

1.4 Scope of Research:

This research focuses on the development of a web-based system capable of recognizing and translating multilingual text from real-world images. The proposed AR-MULTI++ Web system integrates technologies such as Optical Character Recognition (OCR), machine translation, text-to-speech synthesis, and augmented reality-style visual overlay to provide an effective multilingual text understanding platform.

The system processes images containing textual information from various real-world sources such as signboards, documents, and product labels. Through image preprocessing and OCR techniques, the textual content is extracted and translated into a user-selected target language. The translated text is then converted into speech using text-to-speech technology, allowing users to listen to the translated information and improving accessibility.

By implementing and evaluating the proposed framework, the research aims to demonstrate the practical applicability of integrating text recognition, translation, speech synthesis, and visual overlay technologies into a single web-based platform for real-world multilingual communication.

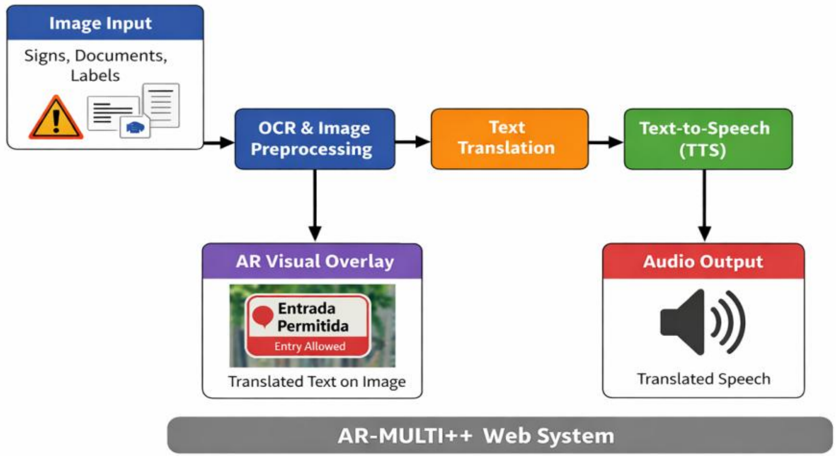

Fig 1.1: Flowchart of AR-MULTI++ Web System for Multilingual Scene Text Recognition and Translation

CHAPTER – 2

LITERATURE REVIEW

The development of multilingual text recognition and translation systems has evolved significantly with the advancement of computer vision, machine learning, and natural language processing technologies. Early text recognition systems relied primarily on traditional Optical Character Recognition (OCR) techniques designed for structured documents such as scanned papers and printed forms [1]. These systems performed effectively when dealing with clear and well-formatted text. However, their performance decreased considerably when applied to natural scene images due to variations in font styles, orientations, lighting conditions, and complex backgrounds. As a result, recognizing text embedded in real-world environments became a challenging research problem.

The integration of computer vision techniques marked a major advancement in scene text recognition. Researchers began developing algorithms capable of detecting and extracting text from images containing natural scenes. Methods such as edge detection, connected component analysis, and texture-based approaches were introduced to improve text localization within images [2]. These techniques helped identify text regions more accurately than traditional document-based OCR methods. Nevertheless, these approaches often struggled with multilingual text recognition and were sensitive to noise and background complexity.

With the emergence of machine learning, more sophisticated methods were introduced to improve text detection and recognition accuracy. Machine learning algorithms such as Support Vector Machines (SVM), Random Forest classifiers, and neural networks were employed to identify text regions and classify characters within images [3]. These models enabled systems to learn patterns from training data, thereby improving recognition accuracy in complex environments. However, the effectiveness of these models depended heavily on handcrafted feature extraction methods, which limited their scalability and adaptability.

The introduction of deep learning technologies revolutionized the field of scene text recognition. Deep learning architectures such as Convolutional Neural Networks (CNNs) demonstrated remarkable success in extracting hierarchical features from images and identifying complex patterns in visual data [4]. CNN-based models significantly improved the accuracy of text detection and recognition in natural scenes. Additionally, advanced neural network architectures enabled systems to process multilingual text and recognize characters from different languages with improved robustness.

Parallel to advancements in OCR technology, machine translation systems also experienced substantial improvements. Early translation methods were based on rule-based and statistical machine translation models, which often produced inaccurate translations due to linguistic complexity and contextual ambiguity. The introduction of neural machine translation models enabled more accurate and context-aware translations by learning semantic relationships between languages [5]. These models allowed translation systems to handle multiple languages and produce more natural translations.

Another important development in multilingual communication systems is the integration of text-to-speech technology. Text-to-speech systems convert textual information into spoken audio, enabling users to listen to translated text rather than only reading it. Modern speech synthesis techniques utilize neural network-based models to generate more natural and intelligible speech outputs [6]. This advancement has significantly improved accessibility, particularly for visually impaired individuals and users who prefer auditory information.

Recent research has also explored the application of augmented reality technologies for visual translation systems. Augmented reality enables translated text to be displayed directly on top of the original image while maintaining spatial alignment with the detected text regions [7]. This approach enhances user understanding by preserving the contextual placement of translated content within the image. AR-based translation systems provide an interactive and intuitive way for users to interpret multilingual information in real-world environments.

Despite these advancements, many existing solutions still operate as isolated systems that perform only one or two functions such as text recognition or translation. Few systems integrate OCR, translation, speech synthesis, and contextual visualization within a single platform. This limitation highlights the need for a comprehensive framework capable of performing multilingual scene text recognition, translation, and audio generation simultaneously. The proposed AR-MULTI++ Web system aims to address this gap by integrating these technologies into a unified web-based platform designed for real-time multilingual communication and improved accessibility.

Related Research Studies

R. Smith – This study focuses on the development of the Tesseract OCR engine for recognizing printed text from scanned documents. The system demonstrates high accuracy for structured text but faces limitations when dealing with natural scene images containing irregular fonts and backgrounds. [1]

A. Mishra et al. – The authors proposed a method for scene text recognition using higher-order language priors and contextual information. The approach improves text detection accuracy in complex images but involves high computational complexity and lacks translation capabilities. [2]

X. Chen et al. – This research presents a comprehensive survey of scene text detection and recognition techniques using deep learning models. The study highlights the evolution of text recognition systems but focuses primarily on theoretical analysis rather than integrated system implementation. [3]

Y. Wu et al. – The research introduces neural machine translation systems capable of translating text between multiple languages using deep learning architectures. Although highly effective for textual translation, the system does not address text extraction from images. [4]

S. Lee et al. – This study proposes an augmented reality-based translation system that overlays translated text on camera-captured images. While it provides contextual visualization, the system remains platform-dependent and lacks integrated speech output. [5]

J. Huang et al. – The authors developed a deep learning framework for multilingual text detection and recognition in natural scene images. The study demonstrates improved performance compared to traditional OCR systems but does not incorporate translation or accessibility features. [6]

M. Chen and Y. Tang – This research combines machine learning techniques with image processing methods to improve character recognition accuracy in complex backgrounds. The approach enhances text detection performance but requires extensive preprocessing. [7]

P.-Y. Brulin et al. – The authors explored deep learning-based models for image text analysis and classification tasks. The study highlights the effectiveness of convolutional neural networks in visual pattern recognition. [8]

H. Lee et al. – This work focuses on integrating speech synthesis technology with translation systems to generate audio output from translated text. The study emphasizes the importance of accessibility in multilingual communication tools. [9]

G. Kaur and P. Chanak – The research introduces intelligent systems that combine image processing and language processing techniques to improve multilingual information access in digital environments. [10]

CHAPTER 3
PROPOSED METHOD

3.1 Methodology

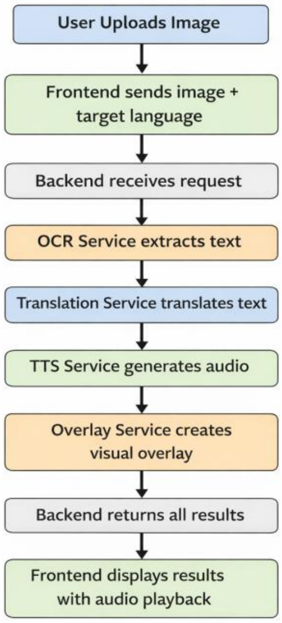

Fig 3.1.1 Block Diagram illustrating the workflow.

3.1.1 Comprehensive Overview of the Research Design:

The proposed research aims to develop a comprehensive framework for recognizing, translating, and presenting multilingual text from real-world images through a web-based platform called AR-MULTI++ Web. The system integrates multiple technologies such as Optical Character Recognition (OCR), machine translation, text-to-speech synthesis, and augmented reality-style visualization to create an intelligent multilingual assistance system. The research methodology is designed in a structured manner, where each stage contributes to the efficient processing and interpretation of textual information present in images.

Initially, the research focuses on acquiring image data containing multilingual text from diverse real-world environments such as signboards, product labels, documents, and public displays. These images represent practical scenarios where users often encounter language barriers. After data acquisition, preprocessing techniques are applied to enhance image quality and improve text detection accuracy. This step ensures that the system can effectively process images containing complex backgrounds, varying lighting conditions, and diverse font styles.

The subsequent stages involve implementing the OCR module to extract textual information from images and identifying the spatial locations of text regions. The extracted text is then passed to the translation module, where machine translation techniques convert the text into the user-selected target language. To further improve accessibility, the translated text is converted into speech using text-to-speech technology. Finally, the translated text is visually overlaid onto the original image in an augmented reality style, preserving the spatial context of the detected text. Through this structured workflow, the system aims to provide an efficient and user-friendly platform for multilingual scene text understanding.

3.1.2 Image Acquisition and Rigorous Preprocessing:

The effectiveness of the AR-MULTI++ Web system relies on the quality and diversity of the input images used during system development and evaluation. Images containing multilingual text are collected from various real-world sources such as public signage, printed documents, educational materials, and product packaging. This diverse dataset ensures that the system is capable of handling multiple languages and different visual conditions.

The preprocessing phase begins with enhancing the quality of the input images to ensure accurate text recognition. Image cleaning techniques such as noise reduction and contrast enhancement are applied to remove unwanted distortions and improve the visibility of textual content. These steps help reduce recognition errors and improve the performance of the OCR module. Following this, normalization is performed to standardize the size and resolution of the images. This ensures consistency in image dimensions and improves the efficiency of subsequent processing modules. Techniques such as resizing and grayscale conversion are commonly used to simplify image processing tasks.

Before performing text recognition, the system identifies regions within the image that potentially contain textual information. Text region detection algorithms analyze patterns within the image to locate areas containing characters or words, and these regions are then passed to the OCR module for further analysis. To enhance system robustness, data augmentation techniques are applied to generate variations of input images. These variations may include rotation, scaling, and brightness adjustments, enabling the system to recognize text under different conditions and improving its generalization capability.

The AR-MULTI++ Web system integrates several computational modules, each designed to perform a specific function within the overall workflow. Optical Character Recognition (OCR) technology is used to detect and extract textual information from images, where the OCR engine identifies characters and words within detected text regions and converts them into machine-readable text. Cloud-based OCR services are utilized to support multiple languages and improve recognition accuracy. The extracted text is then processed by the machine translation module, which converts the source language text into a user-selected target language using neural network-based models to ensure contextually accurate translations.

Once the translation is completed, the translated text is converted into spoken audio using the text-to-speech (TTS) module. This functionality enhances accessibility by allowing users to listen to the translated content, with modern speech synthesis techniques generating clear and natural-sounding audio output. In addition, the augmented reality overlay module displays the translated text directly on the original image while maintaining its spatial alignment with the detected text regions. By preserving the original context of the image, users can easily interpret the translated information.

The overall development of the system involves integrating these modules within a web-based architecture. The frontend interface allows users to upload images and view the results, while the backend server processes the images and performs OCR, translation, speech generation, and overlay creation, ensuring smooth communication between all components of the system.

3.1.4 Rigorous Testing and Validation for System Effectiveness:

The evaluation of the AR-MULTI++ Web system involves assessing its performance across multiple parameters to ensure reliability and effectiveness in real-world scenarios. Performance metrics such as text recognition accuracy, translation quality, processing latency, and audio generation success rate are used to measure system performance.

Text recognition accuracy evaluates how effectively the OCR module extracts characters and words from images. Translation quality is assessed based on semantic correctness and linguistic accuracy. The system’s response time is measured to determine whether it can provide results in real time. Additionally, the clarity and intelligibility of generated speech are evaluated to ensure effective audio output.

Through this detailed evaluation process, the system’s strengths and limitations are identified. The results provide valuable insights into the practical usability of the proposed framework in real-world multilingual communication scenarios. By integrating OCR, translation, speech synthesis, and augmented visualization within a single platform, the AR-MULTI++ Web system aims to provide a robust and accessible solution for overcoming language barriers.

3.2 Implementation

3.2.1 System Processing and Evaluation

The implementation of the AR-MULTI++ Web system represents a carefully structured process aimed at integrating multiple technologies for multilingual text recognition and translation. The system is developed using modern web technologies that combine frontend and backend frameworks to ensure efficient processing and user interaction. The frontend interface is developed using Next.js and React**,** which provide an interactive environment where users can upload images containing textual information. The backend server is implemented using Node.js and Express, which manages image processing, OCR operations, translation requests, and audio generation.

The system workflow begins when a user uploads an image containing multilingual text through the web interface. The uploaded image is transmitted to the backend server where it undergoes preprocessing and analysis. The OCR module plays a central role in extracting textual information from the image. A cloud-based OCR engine is utilized to detect characters and words from the image and convert them into machine-readable text. The OCR module also identifies the spatial coordinates of the detected text regions, which are later used for overlaying translated text on the image.

Following the text extraction process, the translation module processes the extracted text using machine translation techniques. This module automatically detects the source language and converts the extracted text into the target language selected by the user. The translation process ensures that the meaning of the original text is preserved while generating a linguistically accurate output.

After translation, the Text-to-Speech (TTS) module converts the translated text into spoken audio. This component improves system accessibility by enabling users to listen to translated information rather than only reading it. Speech synthesis technologies are used to generate natural and intelligible audio output that corresponds to the translated text.

Another important component of the system is the augmented reality-style overlay module**,** which visually integrates the translated text with the original image. Using the bounding box coordinates obtained from the OCR module, the translated text is placed on top of the original image at the corresponding positions. This overlay mechanism preserves the contextual relationship between the text and its surrounding visual environment.

3.2.1.1 Processing Workflow

The processing workflow represents a crucial stage in the operation of the AR-MULTI++ Web system, where multiple computational modules collaborate to generate meaningful outputs from input images. The workflow begins with image upload and preprocessing, where the system prepares the image for analysis by improving clarity and ensuring compatibility with OCR processing.

The OCR module then analyzes the image and extracts textual information along with its positional coordinates. This information is forwarded to the translation module, which converts the detected text into the target language selected by the user. During this stage, the system ensures that the translated text retains its semantic meaning and contextual relevance.

Subsequently, the translated text is passed to the text-to-speech module to generate audio output. This audio output allows users to listen to the translated information, thereby enhancing accessibility and usability. At the same time, the overlay generation module uses the bounding box information to display the translated text directly on the original image while maintaining spatial alignment.

The entire processing pipeline is designed to operate efficiently, ensuring that users receive results in a short processing time. Through careful integration of OCR, translation, speech synthesis, and overlay generation modules, the system delivers an intuitive and interactive multilingual translation experience.

3.2.1.2 System Evaluation:

Following the implementation phase, the effectiveness of the AR-MULTI++ Web system is evaluated through a detailed analysis of its performance across multiple operational scenarios. The evaluation focuses on measuring the accuracy and efficiency of the system in recognizing multilingual text, translating the extracted content, generating audio output, and overlaying translated text onto the original image. This evaluation stage provides crucial insights into the system's ability to perform reliably in real-world environments where images may contain complex backgrounds, varying lighting conditions, and diverse font styles.

A comprehensive set of performance parameters forms the foundation of this evaluation process. Metrics such as text recognition accuracy, translation quality, processing time, and audio generation reliability are carefully analyzed to determine the overall effectiveness of the system. Text recognition accuracy measures how effectively the Optical Character Recognition module identifies characters and words from images. Translation quality evaluates the semantic correctness and contextual relevance of the translated text. Processing time assesses how quickly the system produces results after receiving an image input, which is important for real-time usability.

Furthermore, the evaluation process extends beyond quantitative metrics to include qualitative analysis of system robustness and adaptability. The system is tested with images containing different languages, varied image resolutions, and diverse environmental conditions to evaluate its stability and consistency. These tests help determine the system’s capability to handle multilingual text recognition and translation tasks across different scenarios.

This comprehensive evaluation ensures that the AR-MULTI++ Web system performs efficiently and reliably in practical applications such as tourism assistance, educational support, and accessibility tools. By systematically analyzing the system’s performance, researchers gain valuable insights into its strengths and potential areas for improvement, thereby contributing to the development of more effective multilingual communication technologies.

3.2.2 Pseudocode for Multilingual Text Recognition and Translation System:

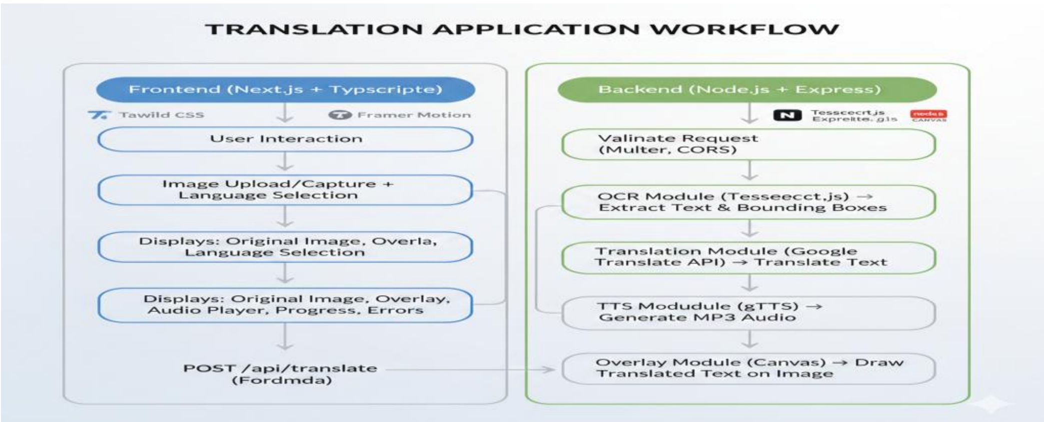

Fig 3.2.2.1: System Architecture

The process begins by importing the necessary libraries required for implementing the system. These include image processing libraries for handling and manipulating images, an OCR library for extracting text from images, a translation API for converting text from one language to another, and a text-to-speech library for generating audio output from translated text.

After loading the required libraries, the web application is initialized. In this stage, the frontend interface is created to allow users to upload images through a web browser, and the backend server is established to process the incoming requests from the user interface.

Once the system is initialized, the image upload and input handling process begins. The application accepts the image input provided by the user through the web interface. The system validates the uploaded file to ensure that the file type and size meet the required specifications. After validation, the image is temporarily stored on the server so that it can be processed by the backend modules.

The next stage involves image preprocessing. During this step, the uploaded image is resized and normalized to ensure consistent processing. Image quality is also enhanced by applying noise reduction and contrast adjustment techniques. These preprocessing operations improve the accuracy of the subsequent text detection and extraction processes.

After preprocessing, the system performs text detection and extraction using the OCR module. The OCR engine analyzes the image to detect textual regions and extract characters and words from the detected areas. The extracted text represents the textual content present in the original image.

Once the text has been extracted, the system proceeds with language detection. In this stage, the source language of the extracted text is automatically identified. Detecting the source language is important for ensuring that the translation process produces accurate results.

Following language detection, the system performs text translation. The extracted text is translated into the target language selected by the user using the translation API. This enables the system to convert multilingual text into a language that the user can understand.

After translation is completed, the text-to-speech generation module converts the translated text into audio output. The generated speech is stored as an audio file on the server so that it can be played through the web interface.

The next step involves augmented overlay creation. The system determines the coordinates of the detected text regions in the image and places the translated text over the original image while maintaining the correct spatial alignment. This creates an augmented reality style overlay where the translated text appears directly on the original scene.

Finally, the system displays the results through the user interface. The application shows both the original image and the overlay image containing the translated text. It also displays the extracted text and the translated text. In addition, the interface provides audio playback so that users can listen to the translated speech.

At the end of the process, the system delivers the final results to the user interface, allowing the user to view the translated image, read the translated text, and listen to the generated audio output.

System Implementation Narrative

In this research endeavor, we implemented the AR-MULTI++ Web system with the objective of enabling multilingual scene text recognition and translation through an interactive web platform. The development process began with the integration of essential software libraries responsible for image processing, optical character recognition, language translation, and speech synthesis. These tools formed the technological backbone required for constructing a reliable and efficient multilingual translation system.

The system workflow commences when a user uploads an image containing textual information. The uploaded image undergoes preprocessing operations aimed at improving visual clarity and preparing the image for text recognition. Techniques such as resizing, normalization, and contrast enhancement ensure that textual regions become more distinguishable for the OCR module.

Following preprocessing, the OCR module analyzes the image to detect and extract textual content. The extracted text is then passed to the translation module, which automatically identifies the source language and converts the text into the desired target language selected by the user. This translation process relies on machine translation algorithms capable of generating contextually accurate language conversions.

Once the translation stage is completed, the system proceeds to generate audio output through the text-to-speech module. This component converts the translated text into spoken language, enabling users to listen to the translated information. The integration of audio functionality significantly improves accessibility, especially for users who prefer auditory interaction.

The final stage of the workflow involves creating an augmented overlay image where the translated text is placed directly over the original image. Using the bounding box coordinates obtained from the OCR stage, the translated text is positioned accurately within the image. This visual overlay maintains the contextual relationship between the text and its surrounding environment, making it easier for users to understand the translated content.

After completing all processing stages, the system displays the results to the user through the web interface. The output includes the original image, the translated overlay image, the extracted and translated text, and the generated audio output. Through this carefully designed workflow, the AR-MULTI++ Web system provides an effective solution for overcoming language barriers in real-world environments.

3.3 Data Preparation

3.3.1 Input Image Processing and Preprocessing

The proposed AR-MULTI++ Web system processes images uploaded by users to recognize and translate multilingual text. Unlike traditional machine learning systems that rely on fixed datasets, this system works with real-time input images captured from real-world environments such as signboards, labels, warnings, menus, and documents. These images may contain text in various languages including Telugu, Hindi, English, and others.

Before text recognition and translation are performed, the uploaded images undergo a preprocessing stage to improve recognition accuracy. This stage ensures that the image quality is suitable for Optical Character Recognition (OCR) processing. The preprocessing operations include image normalization, noise reduction, and resizing to maintain compatibility with the OCR engine.

Fig 3.3.1 Input Image Before Processing

The input image shown above contains a warning sign written in the Telugu language indicating electrical danger. Such real-world images often include graphical symbols, colored backgrounds, and stylized fonts, which can make text recognition challenging.

During preprocessing, the system identifies textual regions within the image and prepares them for recognition using the OCR module. The preprocessing stage improves the clarity of textual content and ensures accurate detection of characters.

After preprocessing, the OCR module extracts the textual information from the image. In this example, the system successfully extracts the Telugu text:

Extracted Text:

"మరణించే ప్రమాదం ఉంది."

The extracted text is then passed to the translation module, which automatically converts the text into the selected target language. In this example, the translated result is:

Translated Text:

"There is danger of dying."

To enhance accessibility and usability, the system further converts the translated text into speech using the Text-to-Speech (TTS) module. This functionality allows users to listen to the translated message, which is especially helpful for individuals with visual impairments or language difficulties.

Fig 3.3.2 Output After Processing

The final output generated by the system includes:

- Extracted text from the image
- Translated text in the selected language
- Augmented overlay displaying translated text on the image
- Audio output for the translated message

The preprocessing and input image processing stage plays a crucial role in ensuring accurate text recognition and translation. By improving image quality and preparing the data for OCR processing, the system effectively extracts multilingual text from real-world images and provides meaningful translations along with audio assistance.

CHAPTER – 4

4.1 Results

The AR-MULTI++ Web system was successfully implemented and tested in a local development environment to evaluate its ability to perform multilingual scene text recognition, translation, and speech generation from real-world images. The system integrates several advanced technologies including Optical Character Recognition (OCR), machine translation, text-to-speech synthesis, and augmented reality style visual overlay. The experimental evaluation demonstrates that the developed system can effectively detect text from images, translate the extracted text into the desired language, and provide an audio output of the translated content. The following sections describe the execution environment, backend processing pipeline, and the user interface results obtained during system testing.

4.1.1 System Execution and Development Environment

The first stage of testing involved running the AR-MULTI++ Web frontend application using the Next.js development server. The application was executed using the command npm run dev, which starts the development server and compiles all necessary frontend modules. As illustrated in Fig 4.1.1, the console output confirms that the system was successfully compiled and initialized. The terminal displays the message indicating that the Next.js application is running locally and accessible through the URL http://localhost:3000. The message “Ready in 5.2s” indicates that the compilation process completed successfully and the frontend interface was launched without errors.

This successful execution confirms that the development environment was correctly configured and all project dependencies were properly installed. The frontend server is responsible for handling user interaction, image uploads, and communication with the backend processing modules. The ability of the application to compile and start successfully demonstrates that the system architecture is stable and capable of supporting further processing tasks such as text extraction and translation.

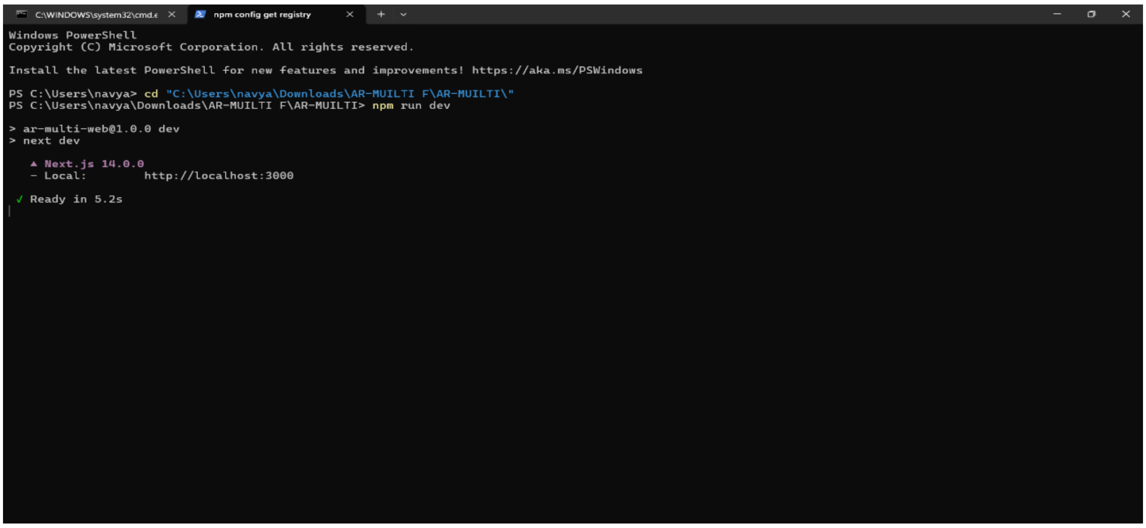

Fig 4.1.1 Running the AR-MULTI++ Web application using the Next.js development server

4.1.2 Backend Server Initialization

After launching the frontend server, the backend server was started to handle image processing operations. The backend application was executed using Node.js along with the nodemon utility, which automatically restarts the server when changes occur in the code. As shown in Fig 4.1.2, the terminal output indicates that the backend server started successfully and is running on port 3001.

The console log also confirms that the Google Cloud Vision API was loaded as the primary OCR engine for extracting text from images. Additionally, the system initializes the uploads directory where processed images and generated audio files are stored. The successful initialization of the backend server indicates that the system is capable of receiving image data from the frontend interface and processing it through the different modules of the AR-MULTI++ framework.

The backend server plays a crucial role in the system architecture because it manages all core processing tasks including OCR text extraction, language translation, speech generation, and overlay creation. The proper initialization of this server ensures smooth communication between the frontend user interface and the processing components.

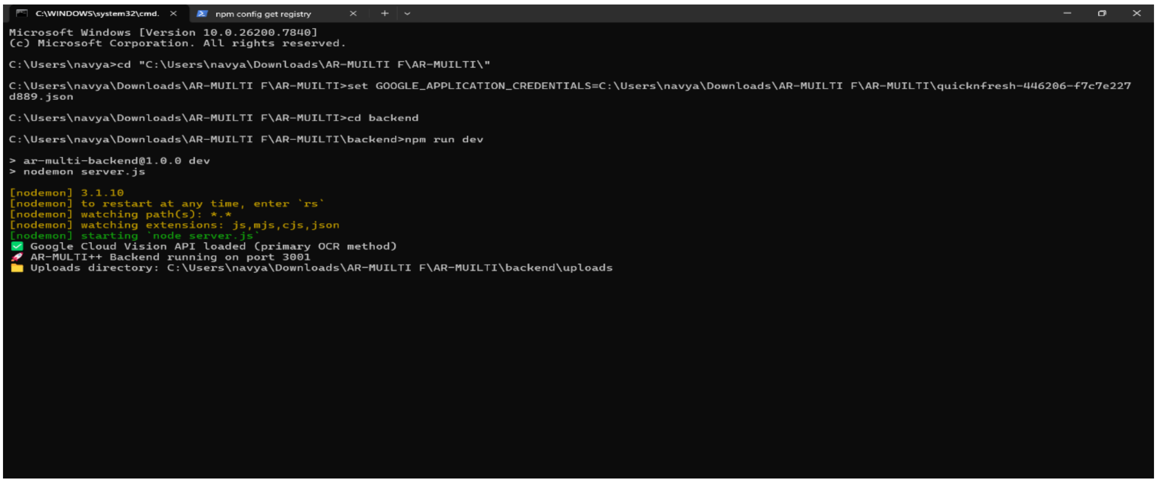

Fig 4.1.2 Backend server initialization and OCR service configuration

4.1.3 Image Processing and OCR Workflow

Once both frontend and backend services were running, the system was tested by uploading an image containing a Telugu warning sign. When the image is submitted through the web interface, the backend server initiates a multi-stage processing pipeline. The detailed terminal output of this processing pipeline is shown in Fig 4.1.3.

The first stage of the processing pipeline involves extracting textual content from the uploaded image using the Google Cloud Vision API. The system begins by reading the image file and sending it to the OCR module. The OCR engine analyzes the image and detects textual elements within it. In this experiment, the system successfully recognized the Telugu sentence "మరణించే ప్రమాదం ఉంది." from the warning sign image. This result demonstrates that the OCR component is capable of recognizing regional language scripts accurately.

After the text extraction stage is completed, the system proceeds to the translation phase. During this stage, the backend server automatically detects the source language of the extracted text and translates it into the target language selected by the user. In this case, the detected language was Telugu, and the text was translated into English. The translated output generated by the system was "There is danger of dying." This translation was performed using the Google Translation API, which provides accurate and context-aware language conversion.

Following the translation stage, the system generates an audio version of the translated text using the text-to-speech module. The translated English sentence is converted into an audio file in MP3 format, allowing users to listen to the translated information. The terminal output confirms that the TTS engine successfully generated an audio file and saved it in the uploads directory of the backend server.

Finally, the system creates a visual overlay by placing the translated text on the original image while maintaining its spatial context. This overlay creates an augmented-reality-like experience where the translated text appears directly on top of the detected text region. This functionality enhances the usability of the system by allowing users to visually understand the translated information while viewing the original image.

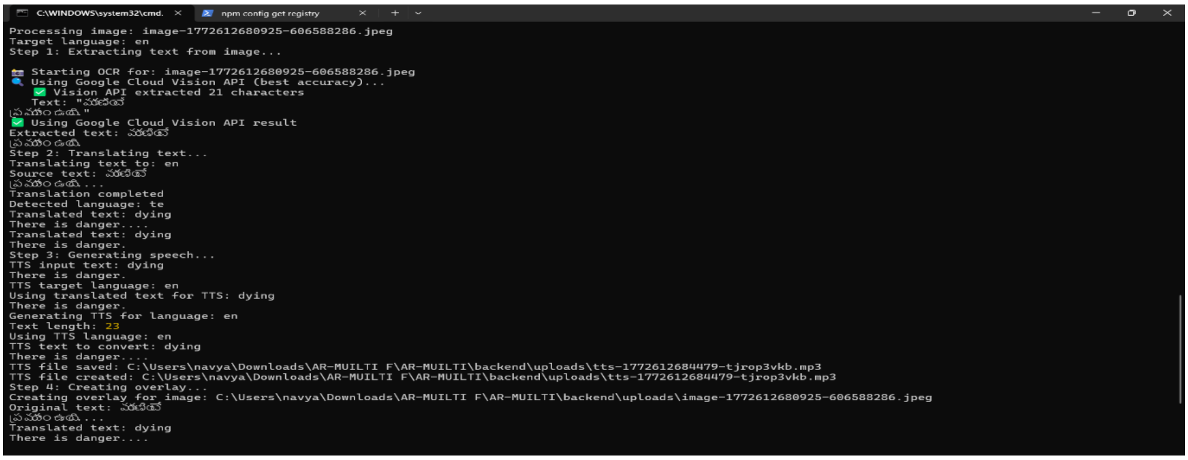

Fig 4.1.3 Backend processing pipeline showing OCR extraction, translation, speech generation, and overlay creation

4.1.4 Web Interface Output and Final Results

The final output of the AR-MULTI++ Web system is displayed through an interactive web interface, as shown in Fig 4.1.4. The user interface clearly presents the results generated by the backend processing pipeline. The interface displays the uploaded image along with the overlay image that contains the translated text. This allows users to visually compare the original content and the translated version simultaneously.

Below the image display section, the system presents the extracted text obtained from the OCR module. In this case, the Telugu text “మరణించే ప్రమాదం ఉంది.” is shown as the detected content from the uploaded image. Immediately after this section, the translated English text “There is danger of dying.” is displayed. This confirms that the translation module successfully converted the extracted text into the desired language.

The interface also includes an audio playback component that allows users to listen to the translated message. By clicking the play button, the user can hear the generated speech output produced by the text-to-speech engine. The interface further provides options to download the generated audio file and the processed overlay image. Additionally, the interface allows users to upload another image for further testing.

The results demonstrate that the AR-MULTI++ Web system successfully integrates OCR, translation, speech synthesis, and visual overlay technologies into a single unified platform. The system effectively processes real-world images, extracts multilingual text, translates the content, and provides both visual and audio outputs. These capabilities confirm that the proposed system can serve as an intelligent multilingual text recognition and translation platform suitable for real-world applications such as travel assistance, safety sign interpretation, and accessibility support.

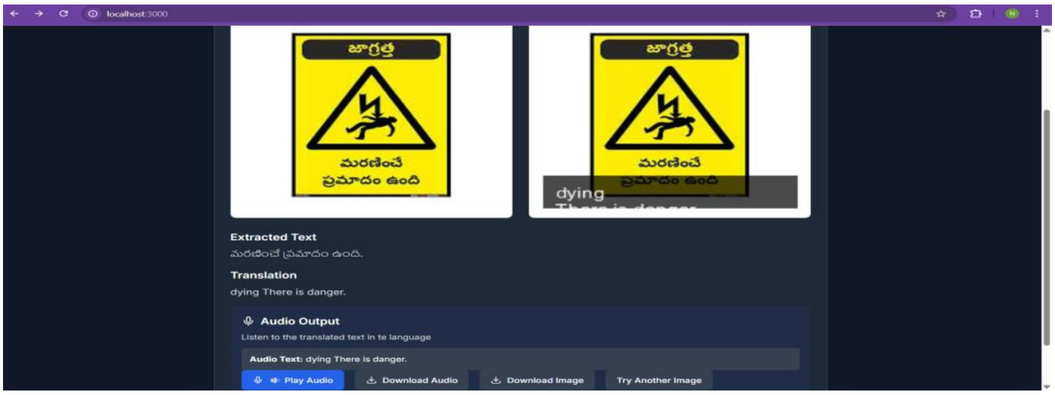

Fig 4.1.4 Web interface displaying extracted text, translated output, and generated audio playback

4.2 Discussion

The implementation and experimental evaluation of the AR-MULTI++ Web system provide valuable insights into the effectiveness of integrating advanced artificial intelligence technologies such as Optical Character Recognition (OCR), machine translation, speech synthesis, and augmented overlay visualization within a unified web platform. The proposed system demonstrates how modern web technologies and cloud-based AI services can be combined to bridge language barriers and enable users to understand multilingual textual information captured from real-world environments. The results obtained from the system confirm that the integration of OCR, translation, and text-to-speech modules can significantly enhance the accessibility and usability of textual information present in images.

The AR-MULTI++ Web system successfully processes images containing multilingual textual content and converts them into meaningful translated outputs accompanied by audio narration. The experimental results indicate that the OCR module effectively detects and extracts text from images using the Google Cloud Vision API, even when the text is embedded within warning boards or signboards. In the conducted experiment, the system accurately identified Telugu text from the uploaded image and converted it into machine-readable text. This demonstrates the ability of the OCR module to handle complex regional language scripts, which is often a challenging task in multilingual image processing applications.

Following the text extraction process, the translation module plays a crucial role in converting the recognized text into the selected target language. The results obtained from the translation process reveal that the system successfully converts the extracted Telugu text into English while preserving the semantic meaning of the original content. The translation output is displayed in the web interface, allowing users to easily understand the meaning of the text present in the image. This functionality is particularly useful for travelers, students, and individuals who frequently encounter unfamiliar languages in real-world environments.

Another significant feature of the system is the integration of the text-to-speech (TTS) module, which converts the translated text into audible speech. The generated audio output enhances accessibility by allowing users to listen to the translated information rather than only reading it. This feature is especially beneficial for visually impaired individuals and users who prefer auditory assistance. The system also allows users to download the generated audio file, which further increases the practicality and usability of the application.

The overlay generation mechanism is one of the most distinctive components of the AR-MULTI++ Web system. Instead of simply presenting the translated text separately, the system overlays the translated content directly onto the image while maintaining visual alignment with the original text location. This creates an augmented reality–like experience, enabling users to view translated information in the same visual context as the original text. The overlay functionality significantly improves the user experience and provides a more intuitive method for understanding multilingual textual content.

The experimental observations also highlight the efficiency of the system's architecture in handling the complete processing pipeline. The system operates through a sequence of stages including image upload, text extraction, translation, speech generation, and overlay rendering. The backend server processes each stage sequentially while maintaining communication with the frontend interface. The system logs displayed in the command terminal confirm that each step of the pipeline is executed successfully, including OCR processing, translation completion, speech generation, and overlay creation.

Furthermore, the graphical web interface developed for the AR-MULTI++ system offers a user-friendly environment that simplifies interaction with the application. Users can upload an image, view the extracted text, read the translated result, and listen to the generated audio output within a single interface. The interface also provides additional functionalities such as downloading the translated audio and downloading the processed image containing the overlay translation. These features collectively enhance the usability and practicality of the system.

Despite its successful implementation, the system exhibits certain limitations that may affect its performance under specific conditions. For instance, the accuracy of text extraction depends heavily on the clarity and quality of the input image. Images containing blurred text, extremely small fonts, or complex backgrounds may reduce the accuracy of the OCR module. Additionally, translation accuracy may vary depending on the linguistic complexity of the extracted text and the contextual limitations of machine translation systems.

Another challenge observed during experimentation involves the precise alignment of translated text in the overlay generation stage. While the current system successfully displays translated text over the image, further improvements can be made to precisely match the exact spatial location of the original text. Advanced computer vision techniques such as bounding box detection and deep learning-based text localization models could be incorporated in future versions of the system to improve overlay accuracy.

Overall, the AR-MULTI++ Web system demonstrates the effectiveness of integrating artificial intelligence technologies into a real-time web application for multilingual text understanding. By combining OCR, translation, text-to-speech, and augmented overlay techniques, the system provides a comprehensive solution for interpreting textual information present in images. The results of this research highlight the potential of such systems to assist users in multilingual environments and improve accessibility to information across language barriers.

4.2.1 OCR Text Extraction Performance

The OCR module represents the foundation of the AR-MULTI++ system, as the accuracy of text recognition directly influences the subsequent translation and speech generation stages. The experimental results show that the Google Cloud Vision API successfully extracts textual content from images containing regional language scripts such as Telugu. The extracted text is displayed in the web interface and logged in the backend terminal, confirming that the OCR module effectively processes the uploaded images.

The system demonstrates strong performance when handling clearly visible text in images such as road signs, warning boards, and printed labels. The OCR module accurately detects characters and converts them into digital text with minimal errors. This capability is particularly important for applications involving real-world image analysis, where textual information must be extracted quickly and reliably.

However, OCR accuracy may decrease when the input image contains distorted text, irregular fonts, or poor lighting conditions. Despite these challenges, the OCR module performs sufficiently well for most practical scenarios and serves as a reliable foundation for the translation and speech generation processes.

4.2.2 Translation and Speech Generation Performance

After extracting text from the image, the translation module converts the recognized text into the selected target language. The experimental results demonstrate that the translation process successfully converts Telugu text into English while maintaining the overall meaning of the sentence. The translated output is displayed clearly within the interface, enabling users to quickly interpret the content of the image.

The text-to-speech module further enhances the functionality of the system by converting the translated text into spoken audio. The generated audio output provides a natural and understandable pronunciation of the translated sentence. Users can play the audio directly within the interface or download the audio file for later use.

The integration of translation and speech synthesis significantly improves the accessibility of the system. It allows users not only to read translated text but also to hear it, making the system useful for individuals who prefer auditory learning or who may have difficulty reading textual information.

4.2.3 System Testing with Real Image Inputs

The AR-MULTI++ Web system was tested using real images containing multilingual textual content to evaluate its performance in practical scenarios. The testing process involved uploading images, extracting text, translating the detected text, generating speech output, and creating an overlay visualization on the original image.

The experimental results confirm that the system successfully completes the entire processing pipeline. The backend logs indicate that each processing stage executes sequentially without errors, including OCR extraction, translation processing, speech generation, and overlay rendering. The frontend interface displays the results clearly, allowing users to interact with the generated outputs.

The successful testing of the system demonstrates its capability to function as a real-time multilingual text interpretation tool. By enabling users to extract, translate, and listen to textual information from images, the AR-MULTI++ Web system provides an effective solution for overcoming language barriers in everyday environments.

CHAPTER – 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This study successfully demonstrated the development and implementation of the AR-MULTI++ Web system, an intelligent web-based platform designed to extract, translate, and interpret multilingual textual content from images. By integrating advanced technologies such as Optical Character Recognition (OCR), machine translation, and text-to-speech synthesis, the system effectively transforms visual textual information into translated and audible outputs. The proposed system highlights the potential of artificial intelligence in simplifying multilingual communication and improving accessibility to textual information present in real-world environments.

The system utilizes the Google Cloud Vision API for accurate text extraction from images containing multilingual content. The extracted text is then processed using machine translation techniques to convert it into the desired target language. Finally, the translated text is converted into speech using text-to-speech technology, enabling users to listen to the translated information. This integrated pipeline demonstrates how multiple AI technologies can be combined to build a comprehensive solution capable of processing and interpreting real-world visual data efficiently.

Experimental results indicate that the AR-MULTI++ Web system performs effectively in detecting and translating textual content from images such as warning boards, signs, and labels. The system successfully identifies regional language text, translates it into English, and generates corresponding audio output. Additionally, the augmented overlay functionality enables users to visualize translated text directly on the original image, creating an enhanced and intuitive user experience similar to augmented reality applications.

The implementation of the AR-MULTI++ Web interface further improves usability by allowing users to upload images, view extracted text, read translated results, and listen to generated audio within a single platform. The system demonstrates how modern web technologies and cloud-based AI services can be integrated to develop practical applications capable of addressing real-world language barriers.

The results obtained from this study confirm that the AR-MULTI++ Web system can serve as a reliable and efficient tool for multilingual text understanding. The successful integration of OCR, translation, speech synthesis, and overlay visualization provides a powerful framework that can assist users in interpreting textual information across different languages. This work contributes to the growing field of intelligent vision-based language processing systems and highlights the potential of AI-driven solutions for enhancing communication and accessibility in multilingual environments.

5.2 Future Scope

Although the AR-MULTI++ Web system demonstrates promising capabilities in multilingual text detection and translation, several opportunities exist for further improvements and extensions. Future research can focus on enhancing the system's accuracy, performance, and real-time capabilities by incorporating more advanced artificial intelligence models and computer vision techniques.

One important area for future development involves improving the text detection and localization process. Currently, the system extracts text using OCR technology, but advanced deep learning models such as YOLO-based text detection or transformer-based OCR models could further improve accuracy in detecting complex or distorted text within images. These enhancements would allow the system to handle challenging scenarios such as curved text, low-light images, or handwritten content more effectively.

Another potential improvement lies in the integration of real-time augmented reality functionality**.** The current system overlays translated text on images after processing; however, future versions could integrate live camera processing**,** enabling users to translate text instantly through their mobile or web camera. This real-time translation capability would significantly enhance the practicality of the system in applications such as tourism, navigation, and public safety.

Future research may also explore expanding the system to support a wider range of languages and dialects**,** particularly low-resource languages that are often underrepresented in machine translation systems. Incorporating multilingual neural translation models could further improve translation quality and contextual understanding across diverse linguistic structures.

Additionally, the system can be extended to include mobile application support**,** allowing users to access the AR-MULTI++ functionality through smartphones and wearable devices such as smart glasses. This extension would create a fully immersive augmented reality experience where translated text appears directly within the user's visual field.

Another promising direction involves integrating context-aware translation and semantic understanding, enabling the system to interpret the meaning of text more accurately based on its surrounding context. This would help improve translation quality, particularly in cases where literal translations may not convey the intended meaning.

Furthermore, future versions of the system may incorporate machine learning-based personalization features, allowing the system to adapt to user preferences, frequently used languages, and personalized translation settings. This would enhance the usability and efficiency of the platform for individual users.

In conclusion, the AR-MULTI++ Web system establishes a strong foundation for intelligent multilingual text interpretation using artificial intelligence and computer vision technologies. Continued research and development in this field have the potential to create more advanced, real-time, and context-aware systems capable of breaking language barriers and facilitating seamless communication across diverse linguistic communities.

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SESHADRI RAO GUDLAVALLERU ENGINEERING COLLEGE

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Seshadri Rao Knowledge Village, Gudlavalleru

Department of Computer Science and Engineering

Program Outcomes (POs)

Engineering Graduates will be able to:

1. **Engineering knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions, component, or software to meet the desired needs.
5. **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. **Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these toone'sown work,as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning**: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

PSO 1: Design, develop, test and maintain reliable software systems and intelligent systems.

PSO 2 : Design and develop web sites, web apps and mobile apps.

PROJECT Cos- POs/PSOs MAPPING

Course Title : B. Tech Capstone Project

Course Code : CS3518

Department : CSE

Batch Number: C7

Academic Year: 2025-2026

PROJECT PROFORMA

Classification of Project	Application	Product	Research	Review
	√			

1. Project Course Outcomes (COs)

CO No	Course Outcome Statement
CO1	Identify and analyze the problem statement using prior technical knowledge in the domain of interest.
CO2	Design and develop engineering solutions to complex problems by employing systematic approach
CO3	Examine ethical, environmental, legal and security issues during project implementation.
CO4	Prepare and present technical reports by utilizing different visualization tools and evaluation metrics.

2. Program Outcomes (POs)

PO No	Program Outcome Statement
PO1	Engineering Knowledge
PO2	Problem Analysis
PO3	Design / Development of Solutions
PO4	Conduct Investigations of Complex Problems
PO5	Engineering Tool Usage
PO6	The Engineer and the World
PO7	Ethics
PO8	Individual and Collaborative Team Work
PO9	Communication
PO10	Project Management and Finance
PO11	Life Long Learning

1. Program Specific Outcomes (PSOs)

PSO No	PSO Statement
PSO1	Design, develop, test and maintain reliable software systems and intelligent systems.
PSO2	Design and develop web sites, web apps and mobile apps

2. CO - PO / PSO Mapping Table (3–High, 2–Moderate, 1–Low)

CS3518 : MAIN PROJECT													
Course Outcomes	Program Outcomes and Program Specific Outcome												
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	P O 11		PS O 1	PS O 2
CO1	3	3	1				2	2	2				1	1
CO2	3	3	3	3	3		2	2	2		1		3	3
CO3	2	2	3	2	2	3	3	2	2	2			3	
CO4	2	2	2		3			3	3	2	2		2	2

Justification of CO-PO/PSO Mapping

CO1: Identify and analyze the problem statement

This outcome strongly maps to **PO1** and **PO2** as students apply fundamental and domain knowledge to understand and define complex problems. It has a low mapping with **PO3** since design activities are minimal at this stage. Moderate mapping is observed with **PO7, PO8**, and **PO9** due to ethical awareness, collaborative discussions, and effective communication during problem identification. It has slight alignment with **PSO1 and PSO2**, as implementation aspects are not yet emphasized.

CO2: Design and develop engineering solutions

This outcome strongly maps to **PO1, PO2, and PO3** as it involves applying knowledge, analyzing problems, and designing appropriate solutions. It also strongly aligns with **PO4** through validation and experimentation, and **PO5** via the use of modern tools and technologies. Moderate mapping is observed with **PO7**, **PO8**, and **PO9** as development requires collaboration and adherence to professional practices. A slight mapping exists with **PO11** as students explore new tools and technologies. This outcome strongly supports **PSO1 and PSO2**, as it focuses on developing reliable software systems and applications.

CO3: Examine ethical, environmental, legal, and security issues

This outcome moderately maps to **PO1** and **PO2** as it involves understanding and analyzing contextual issues. It strongly aligns with **PO3** since solutions must consider constraints such as safety, sustainability, and societal impact. Moderate mapping is observed with **PO4** and **PO5** in evaluating these aspects using appropriate tools and methods. It strongly maps to **PO6** and **PO7** due to emphasis on societal, environmental, legal, and ethical responsibilities. It also moderately contributes to **PO8**, **PO9**, and **PO10** through awareness and discussion of these issues. Strong alignment is observed with **PSO1**, as secure and reliable system development requires these considerations.

CO4: Prepare and present technical reports

This outcome moderately maps to **PO1, PO2, and PO3** as it involves documentation of knowledge, analysis, and design aspects. It strongly aligns with **PO5** through the use of visualization and reporting tools. Strong mapping is observed with **PO8** and **PO9** as effective collaboration and presentation skills are essential. It moderately contributes to **PO10** through structured organization and reporting of project work. It also shows moderate alignment with **PO11** as students enhance their ability to learn and present new concepts. This outcome moderately supports **PSO1 and PSO2**, since documentation and presentation are integral parts of software and application development processes.

5. Sustainable Development Goals (SDG) Alignment

SDG No	Goal	Relevance to the Project
SDG 4	Quality Education	Helps students understand multilingual learning materials by translating text from images into a readable and understandable format
SDG 9	Industry, Innovation & Infrastructure	Uses advanced technologies like OCR, AI, and web-based systems for intelligent image processing and translation
SDG 10	Reduced Inequalities	Reduces language barriers by enabling people from different linguistic backgrounds to access and understand information
SDG 11	Sustainable Cities and Communities	Assists in understanding public signboards, warnings, and instructions in different languages for better urban living
SDG 16	Peace, Justice and Strong Institutions	Helps in interpreting official documents and public notices, improving access to information and transparency

Signature(s) of Students: Signature of Project Supervisor:

1.

2.

3.

4.

from: ICCET SCIE
SCOPUS <iccet2026@gmail.com>

Dear Author/s,
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IN ASSOCIATION WITH
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